

# Phototransduction Mechanism in Retinal Rods and Cones

## The Friedenwald Lecture

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Vision begins in the rod and cone receptors of the retina, where light is absorbed and a neural signal is generated. This signal, in the form of an electrical hyperpolarization of the cell membrane,<sup>1</sup> influences second-order visual neurons by modulating the rate of neurotransmitter (glutamate) release from the synaptic terminal of the photoreceptor. This release is high in the dark and is reduced in a graded fashion by light.<sup>2-9</sup> The response of the postsynaptic neurons to light can be a membrane hyperpolarization or depolarization,<sup>10,11</sup> depending on whether a particular synapse is sign-preserving or sign-inverting.

The overall picture of how light generates a membrane hyperpolarization in rods and cones has been known since the late 1960s.<sup>1</sup> The surface membrane of the photoreceptor outer segment has ion channels—the “light-sensitive” channels—that are open in darkness and are permeable to Na<sup>+</sup>. Driven by its electrochemical gradient, Na<sup>+</sup> steadily enters the outer segment in darkness, constituting a “dark current”<sup>12,13</sup> that partially depolarizes the cell and maintains a high steady release of neurotransmitter from the synaptic terminal. In the light, absorption of photons by the visual pigment in the outer segment somehow leads to the closure of these cation channels,<sup>1</sup> thus stopping the dark current and resulting in a membrane hyperpolarization (Fig. 1).

In rods, the visual pigment is predominantly situated in the membranes of the internalized discs<sup>14-16</sup> within the outer segment. Thus, a diffusible cytoplasmic messenger appears necessary to mediate between the light absorption event and the closure of the light-sensitive channels on the surface membrane.<sup>17</sup> In cones, where the disc membranes and the surface membrane are continuous with each other,<sup>14-16</sup> this

relay via a second messenger is not strictly necessary, but considerations on the high gain of phototransduction also suggest the presence of a second messenger.<sup>18</sup> For many years, two separate ideas on this second messenger had persisted. The first was that light triggers in the photoreceptor outer segment an increase in Ca<sup>2+</sup>, perhaps by release from the membranous discs in rods and by influx across the plasma membrane in cones, and the Ca<sup>2+</sup> acts as a direct blocker of the light-sensitive channels (the Ca<sup>2+</sup> hypothesis<sup>17,19</sup>). The other idea was that in darkness the light-sensitive channels are somehow kept open by the cyclic nucleotide guanosine 3':5'-cyclic monophosphate (cGMP), which is known to be present at high concentration in the outer segment, and that light closes the channels by stimulating the hydrolysis of cGMP (the cGMP hypothesis) (see 20 for an early review). Both ideas had their shares of support and objection from experimental data and were not necessarily mutually exclusive.<sup>21</sup> Variations on these two ideas had also been proposed.<sup>21</sup>

During the past decade, advances in our understanding of this problem have been particularly rapid, (see 22 to 32 for reviews) with important contributions from many laboratories around the world. Here I summarize very briefly experiments done in my own laboratory on this topic. Details of most of the work are already published.<sup>33-55</sup>

The focus here will be on rod receptors, where most of the findings were made. At the end, however, I shall mention some experiments on cones as well. The transduction mechanisms are basically similar in the two kinds of receptors, with mostly quantitative differences between them.

### ION SELECTIVITY OF LIGHT-SENSITIVE CHANNEL IN RODS

Our experiments began with the light-sensitive channel, which has long attracted attention because of its central role in phototransduction. One peculiarity with this channel is that even though the light response shows a reversal potential of 0 to +10 mV,<sup>56-58</sup> therefore suggestive of a relative nonselectivity among cat-

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*Previously supported by USPHS grants EY03553 and EY00222 (RCDA), and by Retina Research Foundation, Houston, Texas. Currently supported by USPHS grant EY06837.*

*The text of this lecture is based in part on a chapter by the author in the book "Cellular and Molecular Biology of the Retina" (Lam DMK, ed., MIT Press, 1988), with permission from the publisher.*

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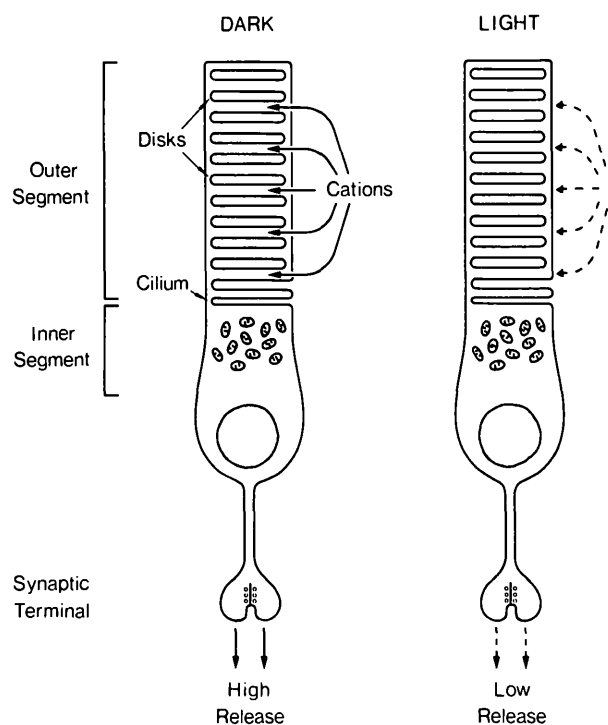


FIGURE 1. Diagram to show the overall response of a rod photoreceptor to light. See text for details.

ions, replacements of external  $\text{Na}^+$  by other monovalent alkali cations invariably removed the dark current and the light response.<sup>59,60</sup> One proposed reconciliation between these two observations<sup>61</sup> was that the channel is not selective for  $\text{Na}^+$  but requires its external presence to stay open.

A strategy adopted by us,<sup>33</sup> and separately by Hodgkin et al<sup>62</sup> to examine this question was to replace rapidly external  $\text{Na}^+$  with other monovalent cations and hopefully to detect a transient inward dark current carried by these other ions before the channel closed due to  $\text{Na}^+$  removal. This strategy indeed worked. Figure 2A shows the behavior of a rod's dark current when all  $\text{Na}^+$  in the external Ringer's solution was rapidly replaced by  $\text{Li}^+$ . The solution exchange time course, shown above the trace, was about 200 ms. It is obvious that the dark current (by convention designated as a negative current) significantly outlasted the removal of  $\text{Na}^+$ , indicating that  $\text{Li}^+$  was indeed able to carry current through the channel. This current, however, was not maintained, but declined to a low level within a few seconds, most probably reflecting the gradual closure of the channels in the absence of external  $\text{Na}^+$  as suggested earlier. Similar observations were made when external  $\text{Na}^+$  was replaced by  $\text{K}^+$ ,  $\text{Rb}^+$  or  $\text{Cs}^+$ . From this kind of experiment, it is possible to obtain an apparent permeability ratio between  $\text{Na}^+$  and the other monovalent cations, essentially by comparing the physiological dark current with that observed *immediately* after cation substitution and there-

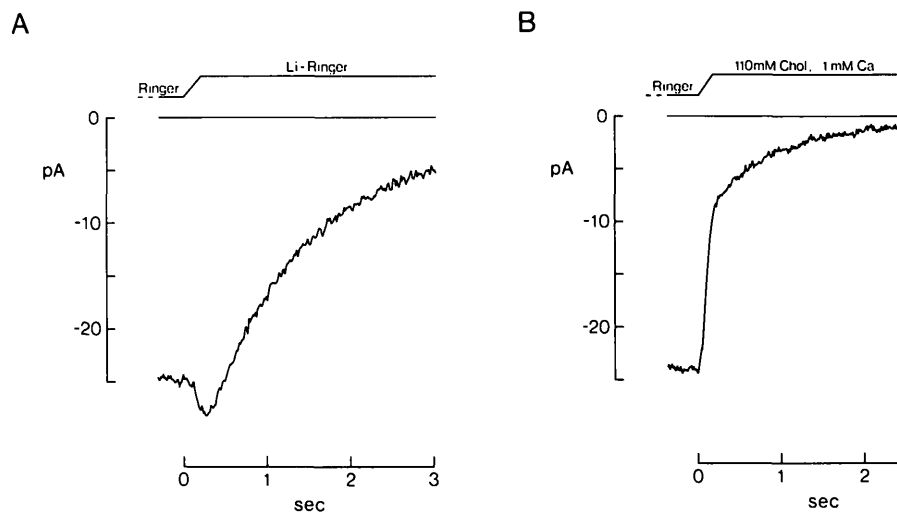
fore before the channel has a chance to close significantly. The permeability ratio between  $\text{K}^+$  and  $\text{Na}^+$  (the two predominant physiological cations) estimated in this way is about 0.7 (33, 62), which can explain the reversal potential of the dark current being near zero mV.

What is perhaps more interesting is that the channel is also permeable to a variety of divalent cations, such as  $\text{Ca}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Ba}^{2+}$ ,  $\text{Sr}^{2+}$ , and so on.<sup>33,39,62-64</sup> Figure 2B shows, for example, an experiment in which all  $\text{Na}^+$ ,  $\text{K}^+$  and  $\text{Mg}^{2+}$  were rapidly removed from normal Ringer's solution (replaced by choline, an impermeant cation), leaving behind  $\text{Ca}^{2+}$  at its physiological concentration of about 1 mM. Again there was a transient residual dark current, undoubtedly carried by  $\text{Ca}^{2+}$ , that outlasted the removal of the other cations. In related experiments,<sup>35,44</sup> we have concluded by indirect estimates that a similar-sized  $\text{Ca}^{2+}$  current is present in the normal dark current, thus ruling out the possibility that the  $\text{Ca}^{2+}$  influx only exists when other permeant cations are removed. In the same manner,<sup>44</sup> we have also observed and quantified a  $\text{Mg}^{2+}$  component in the dark current. Therefore, the dark current through the light-sensitive channel can be broken down into a  $\text{Na}^+$  component (about 80%), a  $\text{Ca}^{2+}$  component (about 15%) and a  $\text{Mg}^{2+}$  component (about 5%). Based on their respective extracellular concentrations, we can calculate<sup>44</sup> their relative probabilities of permeation, given by  $\text{Na}^+:\text{Ca}^{2+}:\text{Mg}^{2+} \sim 1:12:2.5$ . Thus, very surprisingly, the channel actually strongly prefers divalent cations, especially  $\text{Ca}^{2+}$ , over monovalent cations; that the physiological dark current is carried primarily by  $\text{Na}^+$  simply results from this ion being predominant in the external solution. Hodgkin et al<sup>62</sup> have arrived at a similar conclusion from analogous experiments. As will be described below, the  $\text{Ca}^{2+}$  permeability of the channel has a crucial role in phototransduction. On the other hand, any specific function of the  $\text{Mg}^{2+}$  permeability in phototransduction is still unknown at present; it may simply provide a pathway for  $\text{Mg}^{2+}$ , involved in myriad cellular functions, to enter the cell.

In the above experiments, we also found that the decline of the dark current in the absence of external  $\text{Na}^+$  could be avoided if any dark  $\text{Ca}^{2+}$  influx was concomitantly eliminated by removing external  $\text{Ca}^{2+}$  (44). This observation suggests that the closure of the light-sensitive channels in zero external  $\text{Na}^+$  is due to an accumulation of internal  $\text{Ca}^{2+}$  (33, 44, 61, 62).

### THE SODIUM-CALCIUM EXCHANGER AND A LIGHT-INDUCED DECLINE IN INTRACELLULAR CALCIUM IN RODS

The existence of a  $\text{Ca}^{2+}$  component in the physiological dark current implies that in the steady state in darkness there must be a mechanism to remove  $\text{Ca}^{2+}$  from the outer segment. The most likely candidate for this



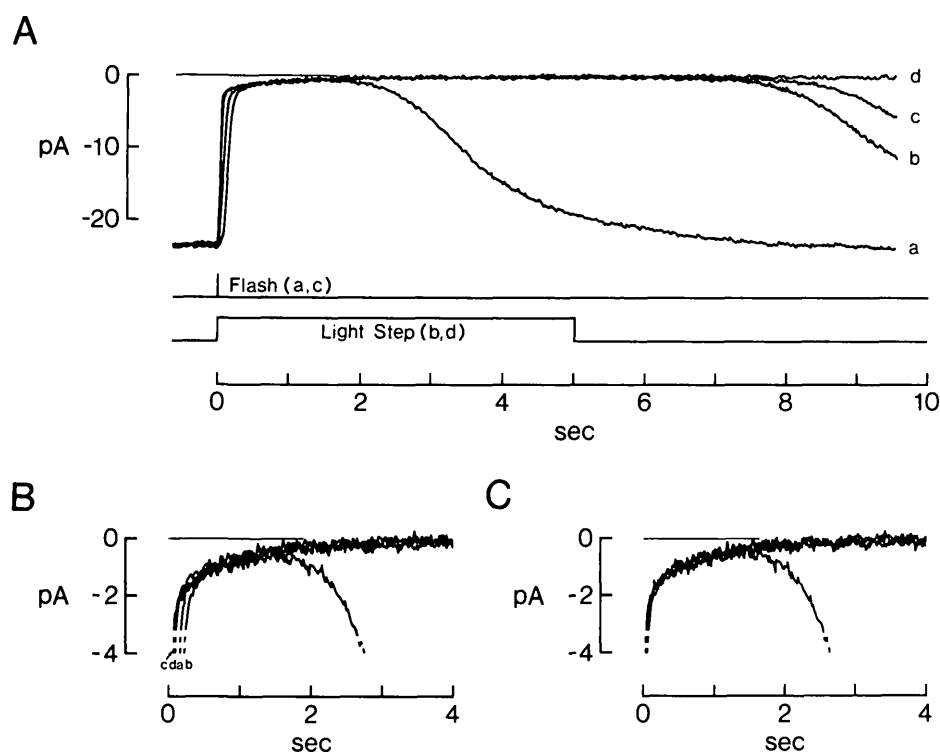
**FIGURE 2.** Experiment to show that the light-sensitive channel in rods is permeable to both monovalent and divalent cations. Top trace in each panel shows the time course of solution change around the outer segment of a toad rod being recorded from. Electrical recording was made by sucking the cell's inner segment and cell body into a tight-fitting glass pipet containing Ringer's solution and connected to a current-recording amplifier.<sup>63,64</sup> Normal Ringer's solution contained (in mM): 110 NaCl, 2.5 KCl, 1.6 MgCl<sub>2</sub>, 1.0 CaCl<sub>2</sub>, 5 TMA (tetramethylammonium)-HEPES, pH 7.6. The lithium Ringer's in **A** contained Li<sup>+</sup> instead of Na<sup>+</sup>, but was otherwise identical to normal Ringer's. The 110 mM Chol-1 mM Ca solution in **B** contained choline chloride instead of NaCl; in addition, KCl and MgCl<sub>2</sub> were omitted, and 0.2 mM TMA-EDTA was added. Reprinted with permission from Lam DMK, ed. *Cellular and Molecular Biology of the Retina*. Cambridge: MIT Press; 1988:41–58.

mechanism is the Na<sup>+</sup>–Ca<sup>2+</sup> exchanger, an Na<sup>+</sup>-dependent Ca<sup>2+</sup> transport mechanism found in photoreceptors<sup>65</sup> and other cell types. Interestingly, for quite a few years this Na<sup>+</sup>–Ca<sup>2+</sup> exchanger had figured importantly in the standing of the Ca<sup>2+</sup> hypothesis. In 1980, it was reported<sup>66,67</sup> that light triggered a net Ca<sup>2+</sup> efflux from the rod outer segment, interpreted at the time to indicate a rise in internal free Ca<sup>2+</sup> which then “spilled” out of the outer segment. This Ca<sup>2+</sup> efflux was shown to be through an Na<sup>+</sup>–Ca<sup>2+</sup> exchange mechanism.

In view of the evidence described earlier for a steady dark Ca<sup>2+</sup> influx, however, we thought the Ca<sup>2+</sup> efflux observed by others during illumination probably deserved a different interpretation. The picture we had<sup>35,44</sup> was simply that in the dark there was a balance between Ca<sup>2+</sup> influx through the light-sensitive channel and Ca<sup>2+</sup> efflux through the Na<sup>+</sup>–Ca<sup>2+</sup> exchanger; in the light the Ca<sup>2+</sup> influx was reduced or stopped, but the Ca<sup>2+</sup> efflux continued, thus leading to a net Ca<sup>2+</sup> efflux. This interpretation, of course, runs against the Ca<sup>2+</sup>-hypothesis because it implies a fall, not a rise, of the internal free Ca<sup>2+</sup> concentration in the light.

One way to settle this dilemma would be to examine the time course of the light-induced Ca<sup>2+</sup> efflux. If the Ca<sup>2+</sup> efflux reflects merely an imbalance between Ca<sup>2+</sup> influx and efflux, the efflux should decline rap-

idly even with maintained illumination because of a steady drain of internal Ca<sup>2+</sup>. On the other hand, if the Ca<sup>2+</sup> efflux reflects a genuine “spilling over” of internal Ca<sup>2+</sup> resulting from a light-induced internal Ca<sup>2+</sup> release, it should be maintained with steady illumination. The experiment of Figure 3 suggests that the first situation is true. Before this experiment is described, it needs pointing out that earlier we have shown<sup>34</sup> that the Na<sup>+</sup>–Ca<sup>2+</sup> exchanger mediating the Ca<sup>2+</sup> efflux is electrogenic; namely, one net positive charge enters the cell for each Ca<sup>2+</sup> ion leaving. Thus, associated with the Ca<sup>2+</sup> extrusion from the outer segment there is a net *inward* membrane current, which we call the exchange current. This exchange current provides a convenient measure of the rate of Ca<sup>2+</sup> efflux. In the experiment of Figure 3, membrane current was recorded from the outer segment of a rod in physiological solution while the cell was stimulated with flashes or steps of light. All of the light stimuli saturated the response; in addition, the flash for trace **c** and the light step for trace **d** were approximately 60 and 110 times brighter than the corresponding ones for traces **a** and **b**. In response to illumination the inward dark current (again designated as a negative current) was rapidly suppressed. The current suppression had two phases: a rapid phase during which most of the dark current disappeared, and a slow phase during which the remaining inward current disappeared (see Fig. 3A). The



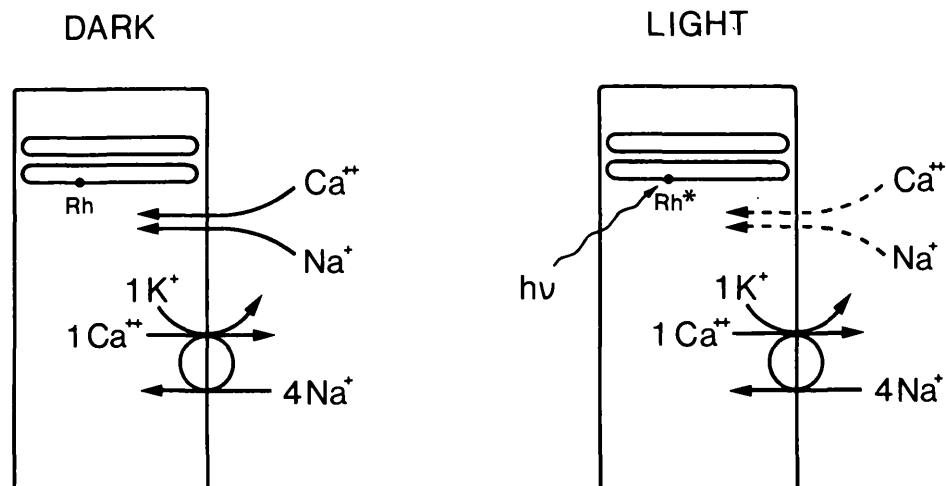
**FIGURE 3.** Responses of a toad rod to bright flashes and steps of light. Similar experimental conditions as in Fig. 2. (A) Traces *a* and *c* indicate flash responses; traces *b* and *d* indicate light-step responses. Flashes *a* and *c* produced about  $4.8 \times 10^2$  and  $5.4 \times 10^4$  photoisomerizations ( $Rh^*$ ), respectively; light steps *b* and *d* produced about  $8.1 \times 10^3$  and  $4.7 \times 10^5$   $Rh^*$   $sec^{-1}$ . Time zero corresponds to the occurrence of flash or the onset of light step. (B) Magnified plots of the secondary rise of the light responses. Same time base as in A. (C) Same traces as in B, but shifted so that the times at half-height ( $t_{1/2}$ ) of the responses coincide. Time zero in this case corresponds to  $t_{1/2}$ . Reprinted with permission from Nakatani K, Yau K-W. *J Physiol.* 1988;395:695–725.

rate of rise of the fast phase increased with increasing light intensity, while the slow phase was insensitive to light intensity. The slow phase turned out to represent the gradual decline of a  $Na^+-Ca^{2+}$  exchange current (i.e., decrease of  $Ca^{2+}$  efflux), because it disappeared when external  $Na^+$  was replaced by  $Li^+$ , which could not drive the exchanger.<sup>35,44</sup> The observation that the light-induced  $Ca^{2+}$  efflux showed both a constant initial amplitude and an immediate decline with a stereotypic time course after onset of suprasaturating light, regardless of the absolute intensity and the duration of illumination (see Fig. 3C), was entirely consistent with the idea that the  $Ca^{2+}$  efflux in the light simply reflected the draining of a pool of internal  $Ca^{2+}$  already present in the outer segment in darkness. Additional support<sup>35,44</sup> to this conclusion is provided by the fact that the initial rate of  $Ca^{2+}$  efflux, i.e., at the onset of bright illumination, is very similar to the steady dark  $Ca^{2+}$  influx rate through the light-sensitive channel, which we separately estimated as mentioned in the previous section.

In short, the above observations implicate the opposite of the  $Ca^{2+}$  hypothesis, namely, a decrease

rather than an increase of the free  $Ca^{2+}$  in the rod outer segment during illumination. Experiments by others,<sup>68–72</sup> including direct  $Ca^{2+}$  measurements with dyes,<sup>69–72</sup> have supported this conclusion. The behavior of the  $Ca^{2+}$  fluxes across the rod outer segment membrane in both darkness and light is summarized in Figure 4. Incidentally, the  $Na^+-Ca^{2+}$  exchanger in rods is now known to involve also  $K^+$  (73, 74); the stoichiometry, as indicated in Figure 4, is  $4Na^+:1Ca^{2+}, 1K^+$ . This more complex stoichiometry allows the transport mechanism to harness the standing electrochemical gradients of both  $Na^+$  and  $K^+$  to extrude  $Ca^{2+}$ , thus allowing a much lower intracellular free  $Ca^{2+}$  concentration to be reached in steady state in the light.<sup>73</sup>

We have also examined the other feature of the original  $Ca^{2+}$  hypothesis, namely, a direct blockage of the light-sensitive channel by cytoplasmic  $Ca^{2+}$ , and have failed to observe it in rod cells as well. For instance, it is possible to load a rod outer segment with excessive amounts of  $Ca^{2+}$  without the channel closing immediately as would be expected from a direct blocking action.<sup>34,62</sup> At the same time, the channel can still



**FIGURE 4.** Behavior of  $\text{Ca}^{2+}$  fluxes at the outer segment plasma membrane in darkness and light. In darkness, there is a balance between  $\text{Ca}^{2+}$  influx through the cGMP-activated channel and an equal  $\text{Ca}^{2+}$  efflux through the  $\text{Na}^{+}\text{-Ca}^{2+}$ ,  $\text{K}^{+}$  exchanger. In the light, the influx is stopped but the efflux continues, thus leading to a decrease in intracellular  $\text{Ca}^{2+}$  concentration. Efflux of  $\text{Na}^{+}$  is via a  $\text{Na}^{+}\text{-K}^{+}$  ATPase at the inner segment. The movement of  $\text{Mg}^{2+}$  is not shown here. Rh, rhodopsin molecule;  $\text{Rh}^{*}$ , photoisomerized rhodopsin molecule;  $h\nu$ , photon. Dashed arrows indicate cessation of cation influx through the light-sensitive channel in the light.

be suppressed by light when the free  $\text{Ca}^{2+}$  concentration in a rod outer segment is buffered.<sup>75</sup> In other words, there is no strict correlation between channel closure and intracellular free  $\text{Ca}^{2+}$  concentration.

It was mentioned earlier that there is also a  $\text{Mg}^{2+}$  component in the inward dark current through the light-sensitive channel. The exit pathway for internal  $\text{Mg}^{2+}$  is not known. We have found this exchanger to transport  $\text{Ca}^{2+}$  and  $\text{Sr}^{2+}$ , but apparently not other divalent cations (unless the transport for these ions is electrically silent).<sup>34</sup>

### ACTIVATION OF THE ROD LIGHT-SENSITIVE CHANNEL BY cGMP

For a long time there had been extensive evidence for the presence of a light-activated enzyme cascade in rods (for review, see 76–78) that led to a powerful hydrolysis of cGMP. Even though its function was uncertain, this cascade had formed the basis of the cGMP hypothesis mentioned earlier. At the same time, electrophysiological evidence<sup>79–82</sup> had indicated that a high level of cGMP in the rod outer segment was linked to the opening of the light-sensitive channel. The nature of this link, however, was likewise unsettled, and often thought to involve  $\text{Ca}^{2+}$ . One idea,<sup>83</sup> for instance, was that a light-induced decrease in cGMP somehow triggered a rise in free  $\text{Ca}^{2+}$ , which then directly closed the channel. Amid all the positive speculations, a major objection against a direct cGMP involvement in phototransduction was that no obvious

reduction in the cGMP content of rod outer segments was detected in the time scale of the electrical response even with bright illumination (for example, see 84, 85). This finding had generated worry even among supporters of the cGMP hypothesis that the cGMP cascade might after all be a phenomenon peripheral to transduction.

With the  $\text{Ca}^{2+}$  hypothesis ruled untenable by the experiments described earlier, however, it was tempting to examine the cGMP idea more closely. In particular, the link between cGMP and the opening of the light-sensitive channel was a central issue. One difficulty with all of the electrophysiological experiments carried out so far was that they were invariably on intact cells. The interpretation of these experiments were not straightforward because any intracellular manipulation, such as an alteration of the cGMP level, was likely to affect other cellular parameters. Thus, any observed change in the degree of opening of the channels could have resulted either directly from a change in the cGMP level or secondarily from changes in other cellular parameters. The only direct way to study the link between cGMP and the channel, it seemed, would be to employ an isolated patch of plasma membrane, which can readily be obtained with the patch-clamp recording technique,<sup>86</sup> so that the conditions around the channel could be strictly controlled. This was a gamble, of course, because the channel could also lose its ability to open in a cell-free situation. Fortunately, the gamble paid off. Fesenko et al<sup>87</sup> first reported the activation of a cation channel in

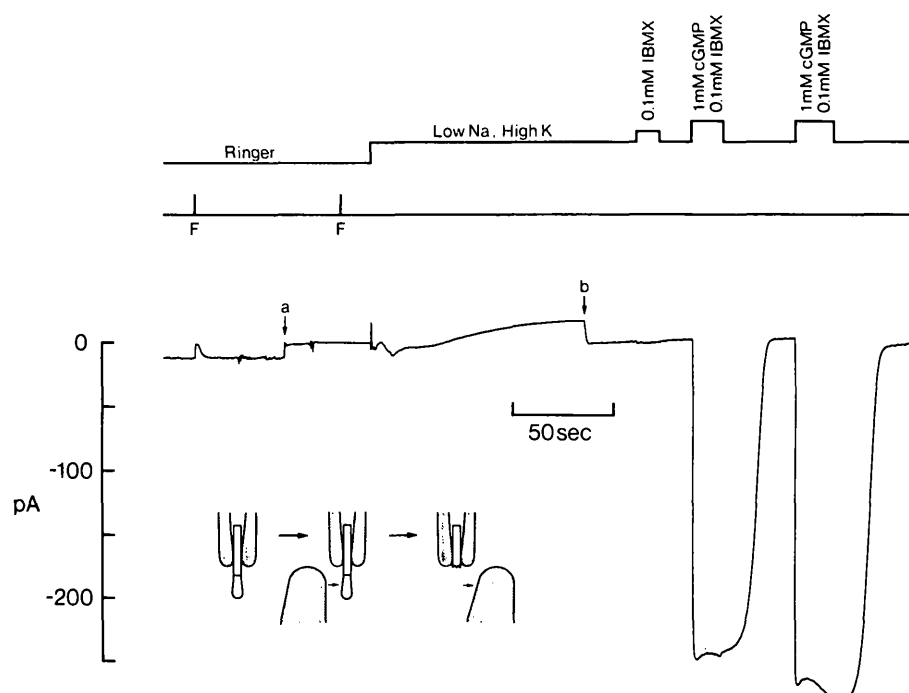
an excised, inside-out patch of rod outer segment plasma membrane when cGMP was applied to its cytoplasmic side. This finding was quickly confirmed by us and others.<sup>37,43,88-90</sup> In addition to providing strong confirmation of the cGMP hypothesis, this remarkable finding is unusual in two respects. One is that it represents the first example of a ligand-activated channel using a cyclic nucleotide as a physiological activator, and the other is that it is a departure from the conventional mode of action of cyclic nucleotides involving kinase stimulation and effector protein phosphorylation. The evidence for the latter point is that ATP, a requisite for protein phosphorylation, is not required for the activation of the channel by cGMP.

It should be pointed out here that earlier experiments based on cation flux measurements already had suggested the existence of a cGMP-activated conductance in partially purified rod disk membrane preparations.<sup>91,92</sup> At the time, with the  $\text{Ca}^{2+}$  hypothesis still very popular, this conductance was thought to regulate  $\text{Ca}^{2+}$  movement in and out of the discs. Because of these prior observations, which continued to receive support from later experiments in other laboratories,<sup>93,94</sup> doubt was raised about the validity of the excised-patch experiment just described. The concern was that the plasma and the disc membranes were anatomically in such close proximity to each other (indeed, observation of structural linkages between the two had been reported<sup>95</sup>) that an excised patch of plasma membrane could easily contain fused fragments of disk membranes. In other words, the cGMP-activated channel might not be present in the plasma membrane at all; instead, the electrical activity observed in an excised patch could have come from contaminating disk membrane fragments. Another point raised about the excised-patch experiment was that any proof of identity between the cGMP-activated channel and the light-sensitive channel should include a demonstration of "light-sensitivity" for the former.

We addressed the above points with a different kind of experiment.<sup>36,45</sup> It is possible to draw most of a rod outer segment into a glass pipette containing Ringer's solution and to shear off the exposed part (Fig. 5 inset). In this way one obtains a truncated, open-ended rod outer segment from which membrane current can be recorded while its interior is dialyzed with cGMP. The strategy is therefore to avoid mechanical disturbance of the plasma membrane over the region of outer segment where membrane current is recorded, so that any possibility of current contribution from disc membrane can be excluded. With such a preparation it was indeed observed, as in an excised patch, that cGMP activated an inward membrane current when present in the dialyzing pseudointracellular solution (Fig. 5). Furthermore, judged from the dose-response relation between activated current and

cGMP concentration (see 36, 45, and Figure 8), this channel appeared to be identical with that observed in an excised-patch experiment. Thus, there is little doubt that the plasma membrane of the outer segment contains a cGMP-activated channel. In the same experiments, we also found that a high concentration (say, 1 mM) of cGMP was able to induce a current (after corrections for membrane potential) as much as 100 times larger than the physiological dark current (see Fig. 5). This result demonstrates that under normal conditions only a very small fraction (about 1%) of the channels in a rod is open in darkness, and, of course, this fraction *never* increases in the light because light only *closes* channels that are open in the dark. It would seem wasteful that the cell makes many more cGMP-activated channels than it actually needs, but there may be a good reason for this.<sup>96</sup> If the cell did get rid of its excess channels, then in order to keep all of the remaining channels open it had to either maintain a much higher concentration of free cGMP in the outer segment or make the affinity of the channel for cGMP much tighter. The first alternative might create problems for cGMP homeostasis in darkness and would also require the phototransduction cascade to work much harder in order to remove the high level of cGMP and close the channels. At the same time, since there is basal cGMP hydrolysis even in darkness, a high steady level of dark cGMP would also lead to a high cGMP metabolic flux, likewise a wasteful situation. As for the second alternative, a tight binding of cGMP to the channel would mean a slow dissociation rate from the channel, resulting in poor time resolution for phototransduction.

From the measured dose-response relation between channel activation and cGMP concentration (see Fig. 8 and section on physiology and structure of cGMP-activated channel), and from the observation that only about 1% of the channels are open in darkness, one can infer that the dark free cGMP concentration in the rod outer segment should be around a few micromolar. The total cGMP concentration in the rod outer segment has been measured biochemically, and is around perhaps 50  $\mu\text{M}$  (see, for example, 84). At the same time, electron microscopy suggests that perhaps only 50% of the rod outer segment volume is free cytosolic space. Putting together these pieces of information, one concludes that only one to a few percent of the total cGMP content in the rod outer segment is free, which is the pool relevant to controlling the channel. This small pool of free cGMP can explain the paradox mentioned earlier that only small changes in cGMP content were found biochemically even with bright illumination. Based on our estimate, even when all free cGMP is hydrolyzed by light and the light-sensitive channels completely close, the change in overall cGMP content in the rod outer segment need only be

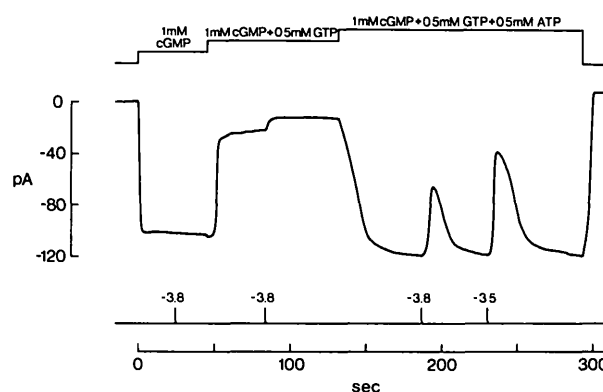


**FIGURE 5.** Recording of the cGMP-induced current from a truncated toad rod outer segment. *a* indicates time at which the outer segment was truncated by the glass probe. *b* indicates voltage adjustment to bring the current baseline back to zero. The cyclic GMP-containing solution had 1 mM cGMP and 100  $\mu$ M IBMX. The IBMX was added to inhibit any cGMP phosphodiesterase still active in the truncated outer segment. Inward membrane current is plotted downward. Uppermost trace indicates the time course of solution change near the pipette tip. Middle trace is light monitor; F, bright light flashes. Reprinted with permission from Nakatani K, Yau K-W. *J Physiol.* 1988;395;731–753.

one to a few percent. More recently, better time-resolved measurements of the light-induced decrease in cGMP in rod outer segments have been carried out.<sup>197–199</sup> These studies suggest that some exchange between free and bound cGMP may also take place in the course of the light response. Where is the majority of cGMP bound to? The location appears to be none other than the cGMP phosphodiesterase in the light-activated cGMP cascade. In addition to the catalytic site that hydrolyzes cGMP, this enzyme has high-affinity, non-catalytic cGMP-binding sites.<sup>97,98</sup> The bound cGMP now appears to have a role in regulating the cGMP phosphodiesterase activity.<sup>99,100</sup>

The question remains: Is the cGMP-activated channel indeed suppressible by light? In the above experiment employing a truncated rod outer segment, the current activated by cGMP was not light sensitive when cGMP was the only metabolite in the dialyzing solution. When GTP was also present, however, the current became light sensitive (Fig. 6). With ATP added to the solution as well, the recovery of the dark current after a flash was significantly accelerated. The first finding thus confirmed the “light sensitivity” of the cGMP-activated channel, provided GTP was present. This GTP requirement is consistent with what is

known about the light-triggered cGMP cascade.<sup>76–78</sup> Briefly, the visual pigment rhodopsin upon absorbing light becomes (in its metarhodopsin II state) catalytically active, and it activates a GTP-binding protein (or simply G protein, also called transducin in retinal pho-



**FIGURE 6.** Effect of light on the cGMP-activated current recorded from a truncated toad rod outer segment. No IBMX was present in any of the solutions. The number above each flash monitor indicates the  $\log_{10}$  attenuation of the light intensity. Reprinted with permission from Nakatani K, Yau K-W. *J Physiol.* 1988;395;731–753.

toreceptors) by promoting an exchange of GTP for GDP bound to the protein at rest. This G protein (from now on called transducin), once activated, in turn activates a cGMP phosphodiesterase to bring about cGMP removal by hydrolysis. Thus, in order for the cascade to function, the presence of GTP is required, as confirmed in the above experiment. The second finding, namely, a "light-quenching" action of ATP, is also consistent with the biochemistry, in that one crucial turn-off mechanism for the light-activated cascade lies in the phosphorylation of the photoexcited rhodopsin by a rhodopsin kinase.<sup>76-78</sup> This covalent modification renders the pigment molecule ineffective in activating any more transducin molecules, with one mechanism involving the inhibitory binding of a 48-kD protein called arrestin or S-antigen to the phosphorylated rhodopsin.<sup>76-78,101</sup> In order for rhodopsin phosphorylation to occur, of course, ATP is required, as demonstrated in Figure 6. Detwiler's group<sup>102,103</sup> has demonstrated similar GTP and ATP requirements in detached rod outer segments.

There was another interesting and ironic twist to the story that is worth mentioning here. As described above, other workers<sup>91-94</sup> had observed a cGMP-induced cation flux in partially purified disc preparations from rod outer segments. More recent evidence,<sup>104,105</sup> however, suggests that this observation in fact resulted from contaminating plasma membrane. In other words, there is a complete turn-around of the picture: the cGMP-activated channel seems to be present *only* in the plasma membrane, and not in the disc membranes at all!

## INTERACTION OF THE ROD CHANNEL WITH DIVALENT CATIONS

It was mentioned in an earlier section that cytoplasmic  $\text{Ca}^{2+}$  does not directly close the light-sensitive channel in an intact rod cell. This does not mean, however, that the channel is insensitive to  $\text{Ca}^{2+}$ , or for that matter,  $\text{Mg}^{2+}$ , another common divalent cation. Indeed, one very surprising observation we made was that under physiological conditions the channel was really subject to continuous, albeit incomplete, blockage by both *external*  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  (38, 41, 43). It appears that in the process of permeating through the channel these cations also obstruct the passage. Matthews<sup>89</sup> has made similar observations. By analogy to the situation with the  $\text{Ca}^{2+}$  channel,<sup>106,107</sup> this phenomenon can be explained by a relatively high affinity of divalent cations for one or more sites within the channel, resulting in long dwell times of these ions in the channel and hence a low rate of ion flux. Simple models on ion permeation and divalent block of the channel have recently been proposed.<sup>108-110</sup>

In physiological conditions, with inward-directed electrochemical gradients for both  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ , the

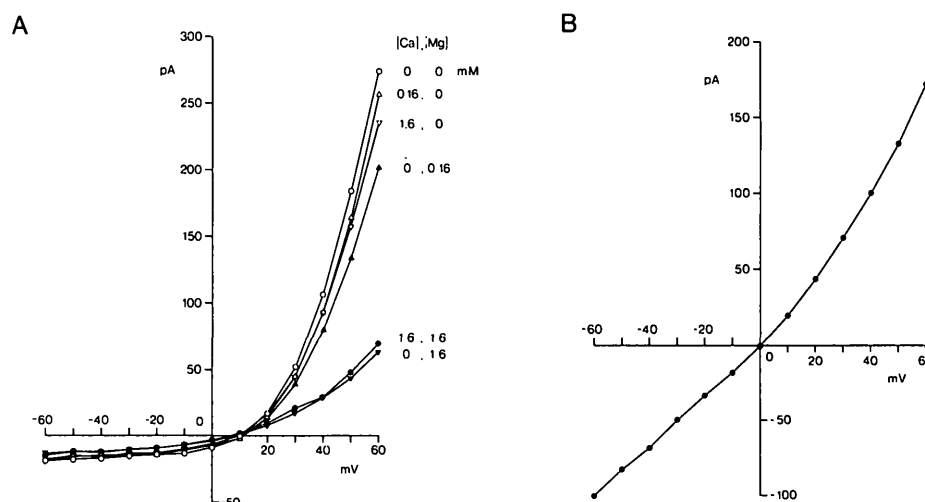
fluxes of these ions are inward; so, according to the above view, the blockage of the channel ought to be determined predominantly by extracellular divalent cations. For the same reason, the channel should be relatively insensitive to divalent cations on the cytoplasmic side, an expectation that can explain the intact cell measurements mentioned earlier. On the other hand, if the polarity of the membrane potential is reversed (i.e., a positive membrane potential, even though this *never* happens physiologically to a rod) the channel should become much more sensitive to divalent cations on the cytoplasmic side. These expectations were borne out experimentally.<sup>41,43</sup> Figure 7A shows the behavior of divalent cation blockage from the cytoplasmic side over a full range of membrane potentials measured in an excised rod membrane patch. In this experiment the extracellular face of the membrane was exposed to normal Ringer's (which contained about 1 mM  $\text{Ca}^{2+}$  and 1.6 mM  $\text{Mg}^{2+}$ ) and the cytoplasmic face was exposed to a pseudointracellular (i.e., low  $\text{Na}^+$  and high  $\text{K}^+$ ) solution containing different concentrations of  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$ . It can be seen that in the negative or hyperpolarizing voltage range (incidentally, the membrane potential of a rod in darkness is about  $-30$  to  $-40$  mV) divalent cations on the cytoplasmic face had little effect on the channel, but in the positive or depolarizing voltage range they showed a much more prominent blocking effect. If divalent cations were omitted from both sides of the membrane, the relation between membrane current and voltage changed dramatically, in that it lost its flatness at negative voltages (Fig. 7B). Thus, the characteristic shape of the current-voltage relation exhibited by the light-sensitive channel (i.e., a flat region at negative potentials and an exponential increase in current at positive potentials) in large part reflects the blocking action of divalent cations as well.<sup>38</sup>

The blockage of the channel by divalent cations serves a subtle but important function. Without the block, the rate of ion flux through the channel would be much higher, giving a prominent single-channel conductance (see next section) and resulting in a relatively high level of background electrical noise associated with random openings of the channels in darkness. With the divalent cation blockage, however, the effective single-channel conductance (after low-pass filtering by the membrane time constant of the cell) becomes much smaller, thus reducing the open-channel noise and improving the signal-to-noise ratio in light detection.<sup>22</sup>

## PHYSIOLOGY AND STRUCTURE OF THE cGMP-ACTIVATED CHANNEL

With excised membrane patches or truncated rod outer segments (see earlier), the dose-response relation between activation of the light-sensitive channel





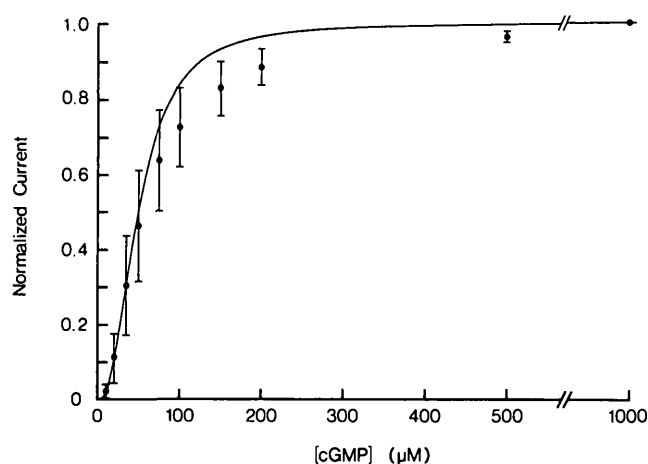
**FIGURE 7.** (A) Effect of divalent cations on the relation between cGMP-activated current and voltage in an outer segment membrane patch excised from a toad rod. Normal Ringer's solution was present on the extracellular side; pseudointracellular solution (in mM: 12.5 NaCl, 100 K gluconate, 1.6  $\text{MgCl}_2$ , 0.1 TMA-EGTA, 5 TMA-HEPES, pH 7.0) was present on the cytoplasmic side except for the indicated variations in  $\text{Ca}^{2+}$  and  $\text{Mg}^{2+}$  concentrations. 1000  $\mu\text{M}$  cGMP. Adjacent points are joined by straight lines. (B) Current-voltage relation for the cGMP-induced current obtained in the absence of divalent cations on both sides of a toad rod membrane patch. Separate patch from A. In this experiment, identical solutions (119 mM NaCl, 5 mM Na-HEPES, pH 7.6) were present on both sides of the membrane, which explains why the reversal potential was at 0 mV. 1000  $\mu\text{M}$  cGMP. Adjacent points are connected by straight lines. Reprinted with permission from Lüttgau H Ch, ed. *Membrane Control of Cellular Activity*. Stuttgart: Gustav Fischer; 1986:343–366.

and cGMP concentration can be measured. From truncated rod experiments,<sup>36,43</sup> for example, this relation has a Hill coefficient ( $n$ ) of 1.8–3.1 (mean  $\sim 2.5$ ) and a half-saturating cGMP concentration ( $K_{1/2}$ ) of 35–85  $\mu\text{M}$  (mean  $\sim 50 \mu\text{M}$ ) (Fig. 8). Experiments with excised membrane patches, which several laboratories have done (see 22 for review), gave values that are broadly similar, though some variations in both  $n$  and  $K_{1/2}$  values have been seen even within experiments from a given laboratory. Some of the variations in  $K_{1/2}$  may depend on the state of phosphorylation of the channel protein.<sup>111</sup> Because the Hill coefficient can be higher than three, the channel must have at least four binding sites for cGMP. Whether the channel can open (and perhaps into one or more other open states<sup>112</sup>) without all the sites occupied by cGMP remains an open question. One unusual feature of the cGMP-activated channel is that, unlike other ligand-activated ion channels, it does not show any desensitization even with prolonged exposure to cGMP.<sup>42,87,113,114</sup> This property is, of course, crucial to phototransduction, by allowing a steady dark membrane current to be maintained and stopped only by light.

Early attempts have been made by others<sup>57,87,115,116</sup> to measure the single-channel conductance of this channel in either intact cells or excised membrane patches. In physiological solutions containing divalent cations, no discrete openings of single channels were detectable, and the single-channel conductance could

only be estimated indirectly with power spectral analysis. This estimated conductance was extremely small, being around 100 fS, raising the question of whether the light-sensitive channel is an aqueous pore at all. Subsequently, however, our laboratory,<sup>42</sup> as well as Zimmerman and Baylor<sup>117</sup> and Matthews and Watanabe,<sup>118</sup> succeeded in observing large single-channel openings activated by cGMP in excised-patch experiments, provided divalent cations were absent. The prominent openings had a conductance of about 25 pS, or about 250 times larger than the 100 fS in physiological solution (Fig. 9). With this large conductance, the cGMP-activated channel is no doubt an aqueous pore like other ion channels. The transformation of the effective single-channel conductance from a large to a minute value by divalent cations serves a function in light detection that was already pointed out in the previous section.

The ability to see single-channel openings also permitted one to examine the gating kinetics of the channel. The individual openings are very brief, being near the limit of recording resolution (Fig. 9). On the average, the individual openings last a millisecond or less, and they appear in clusters or bursts of one to several milliseconds (sometimes longer) in duration.<sup>42</sup> One simple interpretation of the bursts is that they represent repeated transitions of a channel between the open and the closed states while cGMP stays bound. The brief burst durations (about one to a few



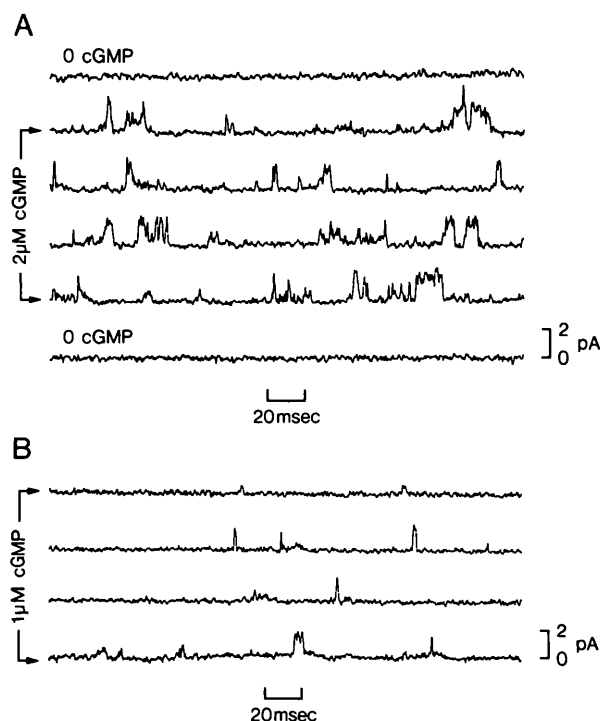
**FIGURE 8.** Dose-response relation between current activation and cGMP concentration. Averaged results from 20 experiments on truncated rod outer segments. Vertical bars indicate standard deviations. The smooth curve is drawn from the Hill equation,  $j_{\text{norm}} = C^n / [C^n + K_{1/2}^n]$ , where  $j_{\text{norm}}$  is normalized current with respect to that at 1 mM cGMP,  $C$  is cGMP concentration,  $n$  is the Hill coefficient (2.4), and  $K_{1/2}$  is the half-saturating cGMP concentration (50  $\mu\text{M}$ ). Reprinted with permission from Nakatani K, Yau K-W. *J Physiol.* 1988;395:731–753.

milliseconds) suggest, then, that cGMP binds to the channel very loosely. Thus, when cytoplasmic cGMP is removed by light, the liganded cGMP should rapidly come off the channel, causing the channel to close. Analysis by Cobbs and Pugh<sup>119</sup> on the rising phase of the rod response has also suggested that the channel can close within a few milliseconds after the onset of bright light. A similar conclusion was drawn by Karpen et al<sup>120</sup> from experiments employing voltage jumps and flash photolysis of caged cGMP. Most recently, Torre et al<sup>121</sup> have reported that the kinetics of the channel openings are even faster than described hitherto. The fast channel kinetics imply that they introduce negligible delay to the overall speed of the light response; the latter is thus dictated by the relatively slow kinetics of the light-activated cGMP biochemical cascade.

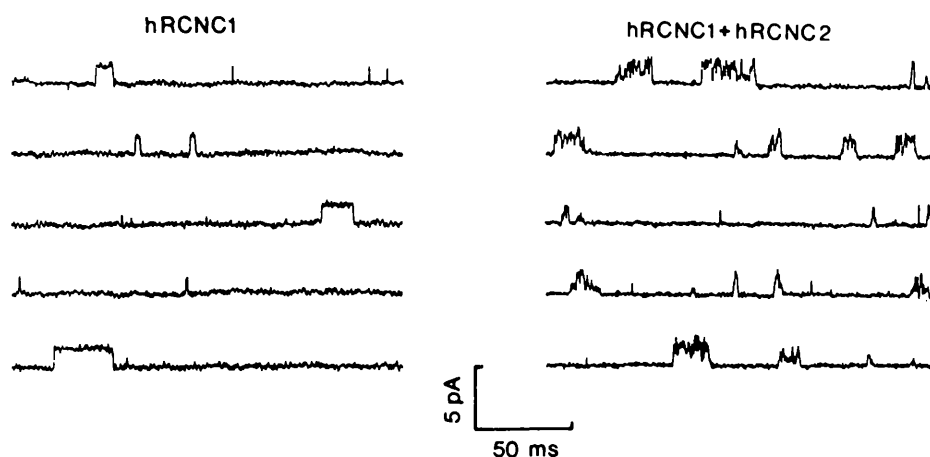
In the late 1980s, Kaupp's group purified a 63-kD protein from bovine retina which, when incorporated into membrane vesicles or a lipid bilayer, was able to produce cGMP-activated channels.<sup>122,123</sup> The cDNA encoding this protein was subsequently cloned which, when expressed in *Xenopus* oocytes, showed the same functional property.<sup>124</sup> Based on its amino acid sequence, a single consensus cyclic nucleotide-binding site was identified,<sup>124</sup> suggesting that the native channel must be a complex composed of at least four copies of this protein. Some homology of the channel protein to voltage-activated  $\text{K}^+$  and  $\text{Ca}^{2+}$  channels has subsequently also been recognized, including similar folding patterns in the membrane as well as the common

presence of an "S4" domain and a pore-forming region.<sup>125–128,200</sup> This new knowledge permits a detailed structure-function study of this channel at the molecular level, which is an active area of research at present. Since the initial discovery of this channel in retinal rods, other ion channels activated or modulated by cyclic nucleotides have now also been identified (see 129 for review). One notable example is the cyclic nucleotide-activated channel mediating olfactory transduction.<sup>130–134</sup>

The finding that the bovine 63-kD protein can form functional cGMP-activated channels by itself has raised the speculation that the native rod channel may be a homo-oligomer, i.e., composed of a single subunit species.<sup>135</sup> Based on the bovine cDNA sequence, we have cloned the cDNA encoding the homologous protein in human retina which likewise forms functional ion channels.<sup>53</sup> In the process, however, we have also cloned a new cDNA species from the human retina that encodes a protein showing only 30% overall homology to the channel-forming protein but nonetheless having a similar folding pattern (based on its hy-



**FIGURE 9.** Openings of individual cGMP-activated channels recorded in the absence of divalent cations from an inside-out membrane patch excised from a toad rod outer segment. Isotonic NaCl solution (118 mM NaCl, 0.1 mM Na-EGTA, 0.1 mM Na-EDTA, 5 mM Na-HEPES, pH 7.6) was present on both sides of the membrane. The holding potential was at +60 mV. All records were low-pass filtered at 2 kHz. **A** and **B** show the channel openings induced by 2  $\mu\text{M}$  and 1  $\mu\text{M}$  cGMP, respectively. Reprinted with permission from Haynes LW, Kay AR, Yau K-W. *Nature.* 1986;321:66–70.



**FIGURE 10.** Openings of single cGMP-activated channels composed of Subunit 1 alone (i.e., homomeric channels, left panel) versus those composed of Subunits 1 and 2 together (i.e., heteromeric channels, right panel). The cDNAs for the two subunits were transfected into HEK 293 cells, and the channel activity was subsequently detected with excised-patch recordings. 20  $\mu$ M cGMP, +60 mV, DC-2 kHz bandwidth. Same solution without divalent cations on both sides of the membrane, containing 140 mM NaCl, 5 mM KCl, 0.5 mM Na-EGTA, 0.5 mM Na-EDTA, 10 mM Na-HEPES, pH 7.6. Note that the long stable openings of the homomeric channels were broken up into bursts of brief openings in the heteromeric channels. See text for details. Reprinted with permission from Chen T-Y, Peng Y-W, Dhallan RS, Ahmed B, Reed RR, Yau K-W. *Nature*. 1993;362:764–767.

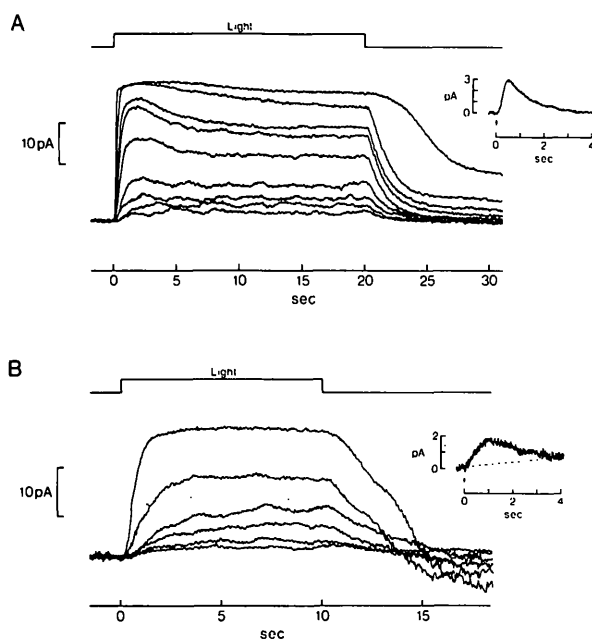
dropathic profile), a cyclic nucleotide-binding domain, an S4 domain, and the pore-forming region.<sup>55</sup> Interestingly, this new protein cannot form functional channels by itself, but when co-expressed with the human homolog of the bovine 63-kD protein, it produces ion channels with kinetic and pharmacological properties very close to those of the native channel.<sup>55</sup> Specifically, the heteromeric channel shows bursts of very brief openings (Fig. 10) as well as a high sensitivity to the blocker *L-cis*-diltiazem,<sup>93</sup> features absent in the homomeric channel.<sup>53,124</sup> Thus, it appears that the native rod channel is made up of at least two dissimilar subunits. We call the original channel-forming protein Subunit 1, and the newly cloned protein Subunit 2 (55). The stoichiometry of these two subunits in the native channel remains to be determined.

### ROLE OF INTRACELLULAR CALCIUM IONS IN TRANSDUCTION

It was stated above that extracellular divalent cations, with  $\text{Ca}^{2+}$  included, exert a continuous block on the light-sensitive channel in darkness. This blockage does not, strictly speaking, participate in the transduction cascade, but instead it provides a subtle improvement of the signalling capability of the rod. As for cytoplasmic divalent cations, it was also concluded that they have a negligible *direct* effect on the channel in normal conditions. The question remained: Does intracellular  $\text{Ca}^{2+}$  *indirectly* affect the channel, thereby having a regulatory role in phototransduction? The answer turned

out to be a resounding yes. There has been longstanding biochemical evidence<sup>136–138</sup> indicating a reciprocal relation between  $\text{Ca}^{2+}$  and cGMP concentrations in the outer segment, i.e., a rise in  $\text{Ca}^{2+}$  concentration suppresses the cGMP concentration, and vice versa. This reciprocity now appears to underlie the phenomenon mentioned earlier of a closure of the light-sensitive channels upon removal of external  $\text{Na}^+$  in darkness, which results in intracellular  $\text{Ca}^{2+}$  accumulation. The mechanism behind this reciprocity is still not entirely clear, but at least part of it appears to arise from an inhibition of guanylate cyclase, the cGMP-synthesizing enzyme, by  $\text{Ca}^{2+}$  (139–143). It is therefore expected that, when light triggers the hydrolysis of cGMP, the resulting decline in  $\text{Ca}^{2+}$  will lead to a disinhibition of the cyclase and hence an increase in cGMP synthesis. This  $\text{Ca}^{2+}$  action produces a negative feedback, with two effects. The first is a speeding of the recovery after illumination. The second, perhaps more important, is that in steady illumination the negative feedback antagonizes the light-induced increase in cGMP hydrolysis and thus down-regulates the sensitivity of a rod to light; in other words, it provides a potential mechanism for light adaptation. This expected  $\text{Ca}^{2+}$  action was indeed confirmed by the experiments of Torre et al.<sup>144</sup>

To more critically assess the importance of this “ $\text{Ca}^{2+}$  feedback” in light adaptation, we have experimentally removed the feedback and examined the effect on light adaptation.<sup>46</sup> The strategy to remove the  $\text{Ca}^{2+}$  feedback consists of removing external  $\text{Ca}^{2+}$  to



**FIGURE 11.** Experiments to show that a negative feedback triggered by a decrease in cytoplasmic  $\text{Ca}^{2+}$  concentration underlies background light adaptation of the rod cell. **(A)** Response-intensity family recorded from a tiger salamander rod in normal Ringer's solution. Note the response relaxation that is indicative of light adaptation. **(B)** Response-intensity family recorded from another salamander rod in a  $\text{ONa}^+-\text{OCa}^{2+}$  solution to remove the  $\text{Ca}^{2+}$  feedback (see text for details). Note the absence of response relaxation in this case. Insets show averaged responses of the two rods to dim flashes under the same conditions as the step responses. Suction-pipet recording method, with cell body and inner segment inside pipet. Reprinted with permission from Nakatani K, Yau K-W. *Nature*. 1988;334:69–71.

eliminate the  $\text{Ca}^{2+}$  influx, and at the same time to replace external  $\text{Na}^+$  with a cation such as  $\text{Li}^+$  or guanidinium, which can carry dark current through the light-sensitive channel but unable to drive the  $\text{Na}^+-\text{Ca}^{2+}$ ,  $\text{K}^+$  exchanger, thus eliminating the  $\text{Ca}^{2+}$  efflux as well (see an earlier section on ion selectivity of the channel). With both  $\text{Ca}^{2+}$  influx and efflux removed, the internal  $\text{Ca}^{2+}$  concentration should no longer depend on the opening or closing of the light-sensitive channel, so the feedback should disappear. Figure 11A shows a control experiment in which a salamander rod bathed in normal Ringer's solution was stimulated with steps of light of different intensities. Note that the response of the cell to a light step rose to a transient peak and then relaxed to a lower plateau level. This response relaxation is a classic manifestation of light adaptation; namely, with continuous illumination, the cell becomes less sensitive to light later on than at the beginning. Figure 11B shows the behavior of another rod in an external solution lacking  $\text{Na}^+$  and  $\text{Ca}^{2+}$  to remove the  $\text{Ca}^{2+}$  feedback. In this case, the response of the cell to a step of light no longer showed

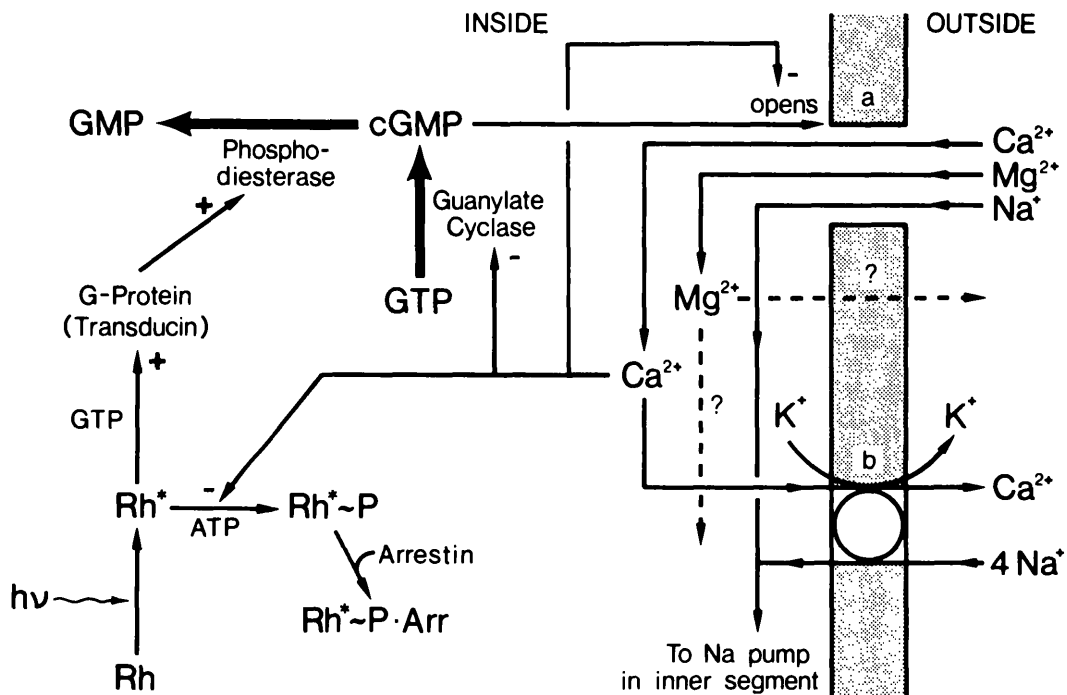
any relaxation, indicating the absence of light adaptation. A more quantitative analysis of these data has led to the same conclusion.<sup>46</sup> In other words, the  $\text{Ca}^{2+}$  feedback triggered by a decline in intracellular  $\text{Ca}^{2+}$  concentration during illumination appears to be the key mechanism underlying background light adaptation. Matthews et al.<sup>145</sup> have arrived at the same conclusion with a similar strategy.

It should be emphasized that, in addition to its function in the light, the  $\text{Ca}^{2+}$  feedback is equally important in darkness. In this case, the negative feedback serves to dampen any spontaneous fluctuations of the cGMP level, which is an inevitable source of dark noise interfering with phototransduction. This dampening not only improves the reliability of light detection, but also guards against any excess  $\text{Na}^+$  influx (which is detrimental to the cell) that might arise from an accidental surge of the cGMP level and consequent opening of more channels.<sup>22</sup>

At present, one direction in the field is to learn more about the biochemical mechanism underlying the  $\text{Ca}^{2+}$  feedback. For example, it is now clear that the guanylate cyclase activity and its dependence on  $\text{Ca}^{2+}$  requires the presence of another protein,<sup>141</sup> possibly a  $\text{Ca}^{2+}$ -binding protein. Recoverin, a recently identified  $\text{Ca}^{2+}$ -binding protein in the retina, was thought for a while to be this protein,<sup>146,147</sup> but no longer,<sup>148,149</sup> and the search continues. It now also appears that  $\text{Ca}^{2+}$  has additional targets in the phototransduction cascade that are important for the negative feedback. Thus, the deactivation of photoexcited rhodopsin via phosphorylation is inhibited by  $\text{Ca}^{2+}$  (150); at the same time, a second  $\text{Ca}^{2+}$  action of unknown molecular mechanism also affects the effective catalytic activity of photoexcited rhodopsin.<sup>151</sup> The  $\text{Ca}^{2+}$  inhibition of rhodopsin phosphorylation does seem to act through the recoverin protein, which now appears to be the same as the protein S-modulin.<sup>150</sup> The result of these modulations on the inactivation of the photoexcited rhodopsin is that, when intracellular  $\text{Ca}^{2+}$  falls in the light, the lifetime of light-activated cGMP phosphodiesterase activity tends to decrease,<sup>148,151–153</sup> amounting to a negative feedback. We have recently succeeded in using the truncated rod outer segment preparation to measure, with electrophysiological recording, the biochemical rate of the light-activated phosphodiesterase activity and its dependence on  $\text{Ca}^{2+}$  concentration.<sup>154</sup> From these measurements, we can quantitatively assess the contributions of the phosphodiesterase and cyclase regulations to the overall  $\text{Ca}^{2+}$  feedback action. Finally, another  $\text{Ca}^{2+}$  target recently identified is, ironically, again the cGMP-activated channel. It appears that the affinity of cGMP for the channel is reduced when a  $\text{Ca}^{2+}$ -calmodulin complex is also bound to the channel.<sup>155</sup> When  $\text{Ca}^{2+}$  falls in the light, calmodulin loses its liganded  $\text{Ca}^{2+}$  and comes

A

## ROD OUTER SEGMENT



B

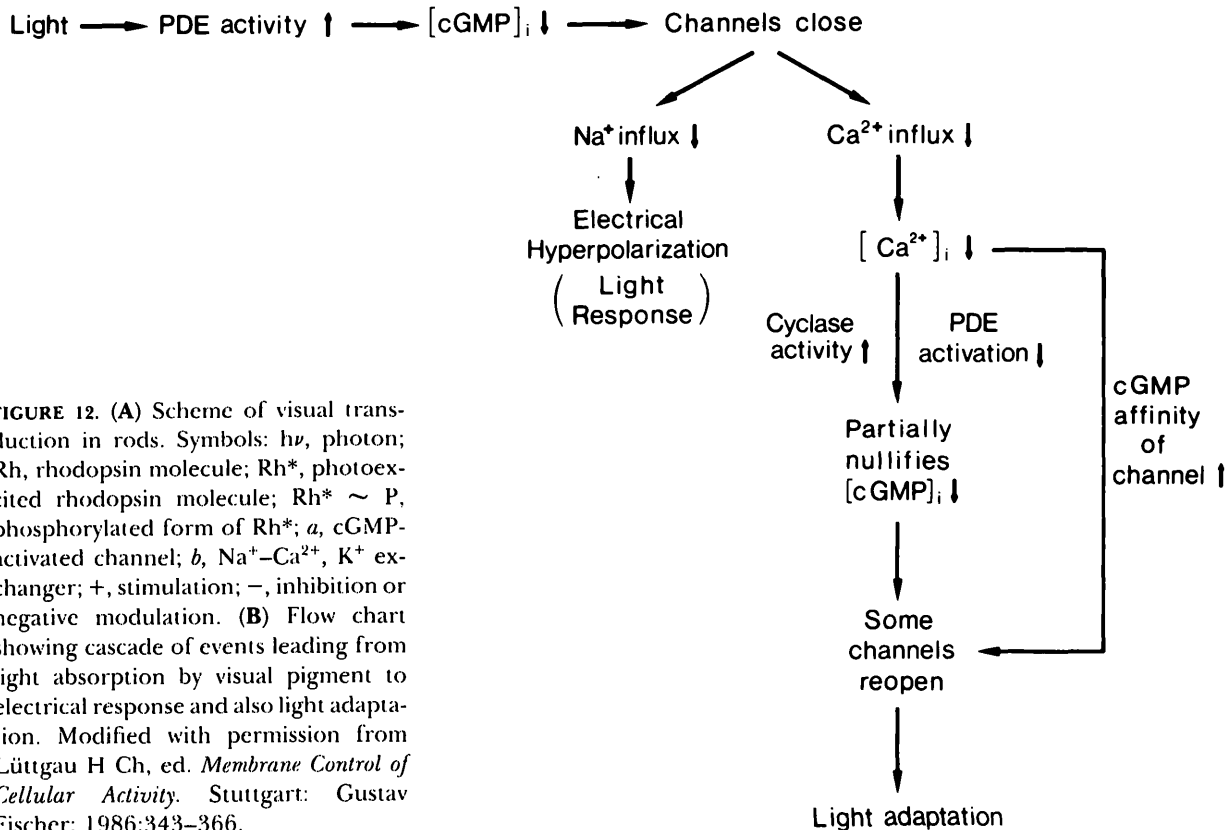


FIGURE 12. (A) Scheme of visual transduction in rods. Symbols:  $h\nu$ , photon; Rh, rhodopsin molecule;  $Rh^*$ , photoexcited rhodopsin molecule;  $Rh^* \sim P$ , phosphorylated form of  $Rh^*$ ; a, cGMP-activated channel; b,  $Na^+ - Ca^{2+}, K^+$  exchanger; +, stimulation; -, inhibition or negative modulation. (B) Flow chart showing cascade of events leading from light absorption by visual pigment to electrical response and also light adaptation. Modified with permission from Lüttgau H Ch, ed. *Membrane Control of Cellular Activity*. Stuttgart: Gustav Fischer; 1986:343-366.

off the channel, rendering the channel more likely to reopen by being more sensitive to cGMP, thus enhancing photorecovery and also leading to light adaptation. Most recently, our group and Robert Molday's group at the University of British Columbia have shown (ARVO Abstracts, 1994) that the  $\text{Ca}^{2+}$  calmodulin binding site resides in Subunit 2 of the channel complex (see section on molecular physiology and structure of the cGMP-activated channel). The  $\text{Ca}^{2+}$ -calmodulin modulation on the channel is not very strong, so its importance in the overall  $\text{Ca}^{2+}$  feedback may only be modest. Finally, there is possibly another protein in the rod outer segment that acts on the channel in a similar manner as calmodulin, but operating at a lower  $\text{Ca}^{2+}$  range;<sup>156</sup> the modulation is again modest, and its physiological significance remains to be determined.

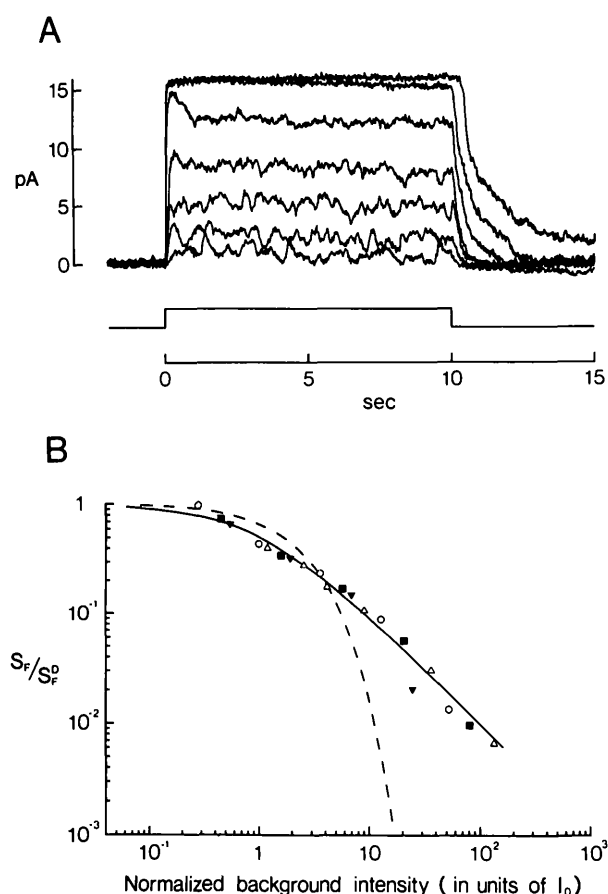
With  $\text{Ca}^{2+}$  after all being so important in phototransduction, a recurrent question in the field is whether the behavior of  $\text{Ca}^{2+}$  in rod outer segments is also influenced by an inositol trisphosphate ( $\text{IP}_3$ )-coupled signalling pathway, which is a widespread  $\text{Ca}^{2+}$ -mobilization mechanism in cells. This question has generated interest because of earlier reports describing a light-activated phosphoinositide pathway in rod outer segments as well as a possible connection between  $\text{IP}_3$  and the rod electrical response to light (see 157 for review). In invertebrate photoreceptors, there is also overwhelming evidence for central roles of phospholipase C (the  $\text{IP}_3$ -generating enzyme) and  $\text{IP}_3$  in phototransduction (see 157–161 for review). So far, however, most electrophysiological investigations have not confirmed the existence of an  $\text{IP}_3$  connection in vertebrate rod phototransduction.<sup>157</sup> Taking advantage of an available polyclonal antibody against the  $\text{IP}_3$  receptor (a  $\text{Ca}^{2+}$ -mobilization channel in the  $\text{IP}_3$  pathway), we have recently looked for the presence of this receptor protein in rod outer segments.<sup>50</sup> We were unable to see any immunostaining in rod (or cone) outer segments, even though clear and consistent staining was seen in the two retinal plexiform layers of different vertebrate species. This result does not support the existence of an  $\text{IP}_3$ -coupled  $\text{Ca}^{2+}$  mobilization pathway in rod outer segments, though it cannot be ruled out that an isoform of the  $\text{IP}_3$  receptor unrecognized by our antibody is present. It is quite possible that the light-activated phosphoinositide pathway in rods is biased toward the diacylglycerol branch of the pathway, which leads to activation of protein kinase C, an enzyme also present in the rod outer segment (see, for example, 162, 163). This area definitely deserves further exploration.

Figure 12A summarizes the key features of the phototransduction process in rods by combining the cGMP cascade and the  $\text{Ca}^{2+}$  actions. Figure 12B shows the series of events triggered by light, leading all the way to light adaptation via the  $\text{Ca}^{2+}$  feedback.

## LIGHT ADAPTATION IN MAMMALIAN RODS

Although the ability of rods of cold-blooded vertebrates to adapt to light has hardly been disputed, cumulative evidence for a long time has pointed against the existence of light adaptation in mammalian rods (see 48 for a brief review). Instead, it was thought that all adaptation in the mammalian retina resides at the "network" level, i.e., postsynaptic to the photoreceptors. When it became clear that light adaptation arose from a negative feedback mechanism acting through  $\text{Ca}^{2+}$ , however, we found the believed difference between lower and higher vertebrate rods rather unsettling. The supposed absence of light adaptation in mammalian rods would suggest the absence of a  $\text{Ca}^{2+}$  feedback in these cells, but this seemed very unlikely because, as pointed out above, the  $\text{Ca}^{2+}$  feedback is also important for cGMP homeostasis and stability of the cGMP-activated membrane current in darkness, properties crucial for the well being of the rod cell. Accordingly, we undertook a fairly comprehensive investigation of light adaptation in rods from a variety of mammalian species, ranging from rats to primates.<sup>48,51,52</sup> We concluded from this study that, counter to previous belief, mammalian rods indeed adapt to light much like, for example, amphibian rods. In both, the dependence of flash sensitivity on background light shows essentially the well-known Weber behavior. Figure 13A shows a family of responses to steps of light elicited from a rabbit rod; the response relaxation indicative of light adaptation is clearly evident. Figure 13B shows collected results from incremental flash-on-background experiments, also on rabbit rods; the data are well fitted by the Weber-Fechner relation. Experiments by others on guinea-pig<sup>164</sup> and human rods<sup>165</sup> have supported our findings. Combining experiments and modelling, we have also shown<sup>52</sup> that the  $\text{Ca}^{2+}$  feedback is present in the phototransduction mechanism of primate rods, and that the feedback can account for the adaptational behavior of these cells fairly well.

Thus, it appears that the functional properties of rods are fairly uniform across all vertebrate species. Our study on primate rods<sup>52</sup> has further indicated that rods of the nocturnal species *Galago garnetti* adapt to light to much the same degree as rods of diurnal species like *Macaca fascicularis* and *Cercopithecus aethiops*, in all cases being broadly similar to rods of lower mammals. This stereotypic behavior suggests that the degree of adaptation in individual rods is perhaps already optimized across species, allowing sufficiently high sensitivity for night vision and at the same time providing Weber behavior (to achieve constancy in contrast sensitivity, an important visual function; see 166) over about 2 log units of background intensity for



**FIGURE 13.** (A) Family of responses to light steps recorded from a rabbit rod. Timing of light step is indicated by upward step below the responses. Suction-pipet recordings; 38°C. Light intensities ranged from 13 to  $3.3 \times 10^4$  photons (500 nm)  $\mu\text{m}^{-2} \text{sec}^{-1}$ . Note the relaxations in the responses. (B) Collected results for the dependence of flash sensitivity ( $S_f$ , normalized with respect to that in darkness, i.e.,  $S_f^0$ ) on background light, also on rabbit rods. Continuous curve is drawn from the Weber-Fechner relation,  $S_f/S_f^0 = [1 + I_s/I_0]^{-1}$ , where  $I_s$  is the steady background light intensity and  $I_0$  is a constant. The dashed curve is drawn according to  $S_f/S_f^0 = \exp[-k_s I_s]$ , where  $k_s$  is a constant. This latter relation is expected for the situation of no adaptation based on a simple model. Reprinted with permission from Nakatani K, Tamura T, Yau K-W. *J Gen Physiol.* 1991;97:413–436.

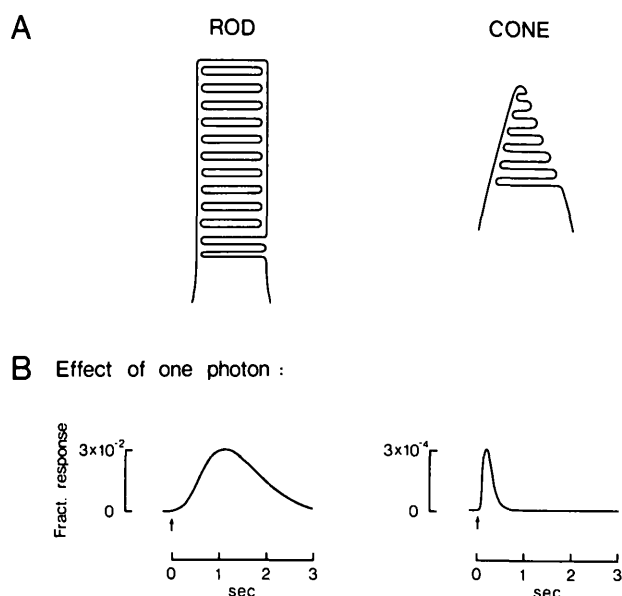
flexibility. Combined with adaptation at the network level, the overall Weber range for the mammalian rod system is reasonably broad, spanning 5 log units of background light intensities. There is a trade-off between sensitivity and dynamic range. A weaker  $\text{Ca}^{2+}$  feedback would increase sensitivity in darkness and dim background light, but also reduce the dynamic range because response saturation would rapidly be reached. A stronger feedback, on the other hand, could extend the dynamic range, but at the expense of sensitivity at low light levels. The rods appeared to have evolved into a compromise between these two

requirements. Thus, to achieve an even wider dynamic range, different nocturnal and diurnal animals have acquired a cone system of varied importance instead of modifying the existing properties of the rod system.

## CONE TRANSDUCTION

Certain differences between rods and cones are worth noting, apart from the fact that they have different visual pigments.<sup>167</sup> Anatomically, the cone outer segment is generally smaller than the rod outer segment. More significantly, however, the membranous discs in the cone outer segment are continuous with the plasma membrane to form a highly convoluted surface membrane<sup>14–16</sup> (Fig. 14A). Thus even though the cone outer segment is much smaller than the rod outer segment, its surface area is actually many times larger. Physiologically, rods and cones are also different in two important ways (see Fig. 14B); namely, cones are much less sensitive to light than rods (typically by 25 to 100 times) and their responses have faster waveforms (typically by several times) (see, for example, 47, 166, 169). The study of cone transduction therefore has to deal with not only its qualitative features, but also its quantitative differences from rod transduction.

We shall first describe the qualitative features of cone transduction. Progress in the understanding of rod transduction has prompted similar experiments on cones. Indeed, by applying patch-clamp recording to a patch of plasma membrane excised from a cone outer segment<sup>38</sup> we were able to show that the light-sensitive channel in cones is likewise directly gated by cGMP. Measurements from intact cells by Cobbs et al<sup>170</sup> have led to a similar conclusion. The characteristics of activation of this channel by cGMP are remarkably similar to those for the rod channel.<sup>38,49</sup> In the absence of divalent cations, the current-voltage relation for the cone channel also resembles that for the rod channel. On the other hand, in physiological ionic conditions, the rod and cone channels show quite different current-voltage relations (Fig. 15), suggesting that the two channels are distinct entities.<sup>38,171</sup> Recent molecular cloning from chicken retina has confirmed that the rod and cone channels are encoded by different genes.<sup>172</sup> Because the characteristic shape of the current-voltage relation comes from interactions of divalent cations with the channel (see earlier description of the rod channel), the downward swing of this relation for the cone channel at negative voltages suggests that permeating divalent cations do not bind as tightly to the pore of the cone channel as to that of the rod channel. Another interesting finding is that, although the cone outer segment has a much larger surface area than the rod outer segment, the density of channels seems to be scaled down proportionally in cones<sup>49</sup> so that the total number is probably comparable in both types of cells.



**FIGURE 14.** (A) Structural differences between the rod and cone outer segments. (B) Physiological differences between the two kinds of receptors. The two waveforms illustrate and compare the responses of rods and cones to a single photon, with amplitudes expressed as a fraction of maximum. The time courses depicted for the responses are typical of amphibian rods and cones at about 20°C. The arrows indicate timings of photon absorption in relation to the response. Reprinted with permission from Lam DMK, ed. *Cellular and Molecular Biology of the Retina*. Cambridge: MIT Press; 1988:41–58.

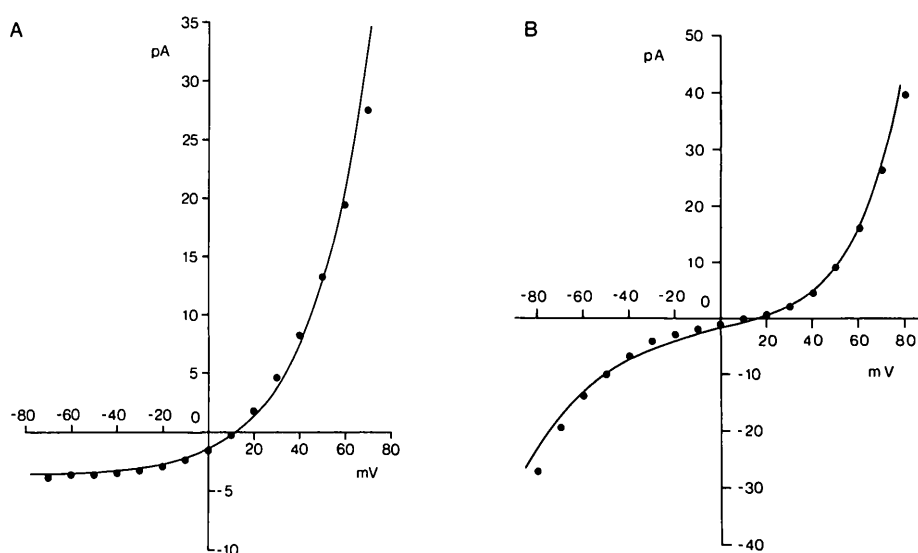
Employing a truncated cone cell preparation<sup>40</sup> similar to the rod preparation described earlier, we were also able to show that GTP was required in order for light to suppress the cGMP-activated channel, implicating the involvement of a G protein (transducin) in cone phototransduction. In the same preparation we also found that ATP quenched the light action, suggesting that phosphorylation of the cone pigment is required for terminating transduction, again as in rods. Most recently, a transducin and a cGMP phosphodiesterase specific to cones have been identified;<sup>174–176</sup> these are likely the intermediates mediating cone transduction. A  $\text{Ca}^{2+}$ -binding protein named visinin has likewise been identified in the chicken retina;<sup>177</sup> from its amino acid sequence, this is probably the cone homolog of the rod recoverin protein. It is thus remarkable that practically every protein along the phototransduction pathway has different isoforms in rods and cones. We have found this to be the case<sup>54</sup> even for the  $\beta$  and  $\gamma$  subunits of transducin, which serve more subtle functions than the  $\alpha$  subunit (which activates the cGMP phosphodiesterase) in signal transduction. Finally, again like rods, only a tiny fraction (about 1%) of the cone channels are open in darkness,<sup>40,170</sup> being maintained by a free cGMP concentration of about a few micromolars in the cytoplasm.<sup>40</sup>

Considering the close parallel between rod and cone transduction, it is no surprise that in cones there is also a balance between a dark  $\text{Ca}^{2+}$  influx through the channel and a dark  $\text{Ca}^{2+}$  efflux through a  $\text{Na}^+$ - $\text{Ca}^{2+}$  exchanger.<sup>47</sup> There is some evidence<sup>173</sup> that the percentage of dark current carried by  $\text{Ca}^{2+}$  is higher in cones than in rods, which may be connected with the distinct current-voltage relation for the cone channel described earlier, but its functional significance is unclear. As far as we can tell,<sup>47</sup> the cone exchanger seems to operate in much the same way as in rods, though whether, for example, it also involves  $\text{K}^+$  transport remains to be checked. Interesting enough, the density of the exchanger on the cone outer segment membrane, like that of the cGMP-activated channels mentioned earlier, seems to be scaled down so that the total number of exchange sites is comparable to that in rods.<sup>47</sup>

In the light, with the  $\text{Ca}^{2+}$  influx through the light-sensitive channel being suppressed, the free  $\text{Ca}^{2+}$  concentration in the cone outer segment should be pumped down by the exchanger, as in rods. When we examined this  $\text{Ca}^{2+}$  decline in cones we found that its rate was much faster than in rods.<sup>47</sup> Others<sup>172,178</sup> have found the same. We were especially interested in this faster  $\text{Ca}^{2+}$  decline in cones because it could potentially explain the differences in light sensitivity and response kinetics between rods and cones. The reason is that since the  $\text{Ca}^{2+}$  decline provides negative feedback to the phototransduction cascade, a faster  $\text{Ca}^{2+}$  decline in cones should allow the negative feedback to cut into the transduction process more rapidly and effectively, thereby reducing both the amplitude and the time course of the light response. Despite its attractiveness, however, we ruled out this idea after comparing the light sensitivities of rods and cones in the absence of external  $\text{Na}^+$  and  $\text{Ca}^{2+}$ , which eliminated the  $\text{Ca}^{2+}$  feedback.<sup>47</sup> We found that, even under these conditions, rods and cones still showed the same degrees of difference in absolute light sensitivity and response kinetics as in control conditions (Fig. 16).

The true mechanisms underlying the differences in transduction gain and kinetics between the two kinds of receptors probably reside in the biochemistry of the cGMP cascade. Figure 17 is another schematic representation (see also 179) of the phototransduction cascade, with the negative feedback mediated by  $\text{Ca}^{2+}$  excluded. The various active intermediates in the cascade (i.e., the photoexcited visual pigment, the activated G protein and the activated phosphodiesterase, marked by asterisks) have characteristic decay time constants designated by  $\tau_1$ ,  $\tau_2$  and  $\tau_3$ . In addition, the dark metabolic flux of cGMP also introduces another integrating time constant,  $\tau_4$ , to the cascade. The overall gain of the cascade, and hence the light sensitivity, depends on these time constants in a *multiplicative*





**FIGURE 15.** Comparison between the current-voltage relations for the rod (A) and cone (B) cGMP-activated channels, obtained with inside-out membrane patches excised from toad rods and catfish cones. In both cases, physiological solution containing divalent cations was on the extracellular side, and pseudointracellular solution on the cytoplasmic side, of the membrane (see Figure 7 legend).

manner (47; see also 180, 181); the longer the time constants (or the slower the decay rates and the dark metabolic flux), the higher will be the sensitivity. Thus, if one assumes that the time constants for the dark metabolic flux and the decays of the active intermediates in cone transduction are uniformly reduced (i.e., becoming faster) by several fold, then one can easily explain the several-fold faster cone response kinetics as well as the 25- to 100-fold lower sensitivity of cones. With the view that the biochemistry of transduction indeed proceeds faster in cones, it is then easy to see why the  $\text{Ca}^{2+}$  feedback in cones should be accelerated as well, in that the kinetics of transduction should match the kinetics of the  $\text{Ca}^{2+}$  feedback in order for the two to work in unison.<sup>47</sup> Indeed, as in rods, the  $\text{Ca}^{2+}$  feedback does seem to be a key mechanism in background light adaptation,<sup>46,182</sup> as well as in bleaching light adaptation.<sup>183</sup>

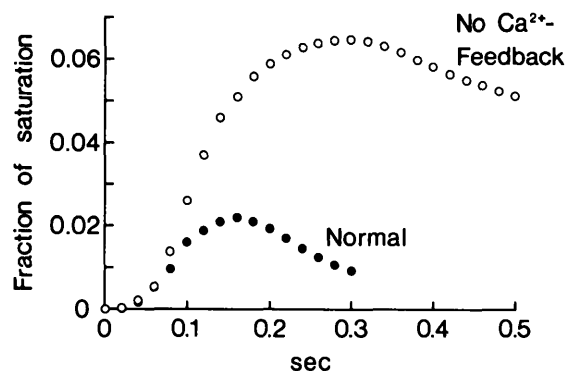
One question no doubt interesting to many people is why the membranous discs in the cone outer segment, unlike their completely internalized counterpart in the rod outer segment, are continuous with the plasma membrane. It is not obvious that the cone outer segment needs a larger surface area, especially considering that the densities of the cGMP-activated channels and the  $\text{Na}^+$ - $\text{Ca}^{2+}$  exchanger are in fact lower in cones, apparently to compensate for this increase in area (see above). One suggestion<sup>184</sup> is that whereas one photon can influence a region spanning many discs in the rod outer segment,<sup>12,185,186</sup> the geometric shape of the cone outer segment restricts the influence of a photon to within one disc and therefore

effectively scales down the light sensitivity of cones. Though interesting, this idea nonetheless does not explain why one photon when absorbed by a cone does not always saturate even one disc, as we have found in tiger salamander cones. Furthermore, anatomic studies<sup>15,187</sup> have indicated that cones in certain animal species (e.g., mammals) have a significant fraction of their outer segment discs completely internalized. Thus, it appears that the sensitivity difference between rods and cones results not so much from outer segment geometry as from a genuine difference in the biochemistry underlying transduction. As to the real purpose of the geometry of the cone outer segment, a simple explanation may be that since cones generally function in bright light and therefore require a rapid recycling of their pigment molecules, the continuum between their disc membranes and the plasma membrane provides rapid access of the regenerated chromophore from the cell exterior to the bleached pigment.

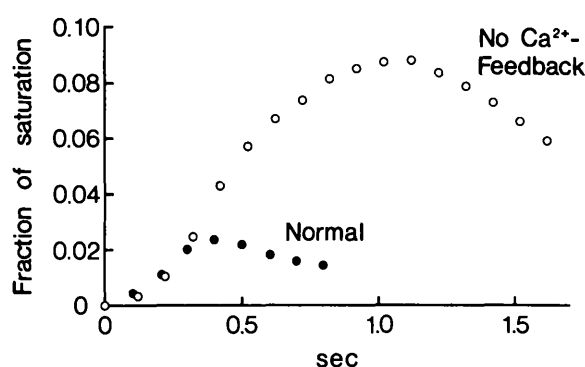
## CONCLUDING REMARKS

The past decade has been an exciting period for the study of visual transduction, with great advances arising from a synergy between biochemical and electrophysiological approaches. In a way, phototransduction is unique in that its triggering stimulus consists of electromagnetic radiation. As more and more is known, however, a remarkable commonality emerges between this process and other signal transduction pathways mediated by G proteins, now known to be

## Cone:



## Rod:

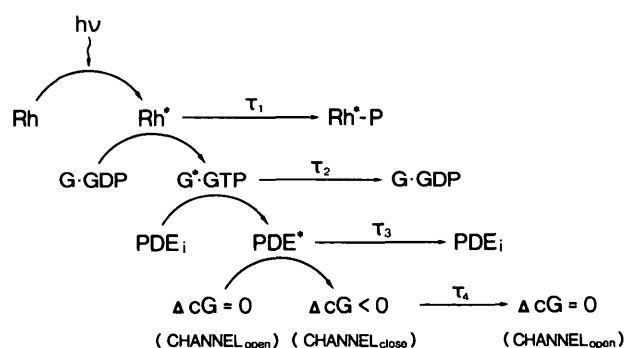


**FIGURE 16.** Experiments demonstrating that the  $\text{Ca}^{2+}$  feedback does not underlie the differences in sensitivity and response kinetics between rods and cones. Suction-pipet recordings. In both top and bottom panels, filled circles indicate an electrical response to a dim flash in normal Ringer's solution, and open circles indicate the response to an identical flash with the  $\text{Ca}^{2+}$  feedback removed. For both rods and cones, removing the negative feedback increases the light sensitivity by threefold to fourfold and slows the time-to-peak of the response, but the same differences between rods and cones persist (note the different scales in top and bottom panels). Flash delivered 85 photons ( $620 \text{ nm}$ )  $\mu\text{m}^{-2}$  in upper panel and 2.2 photons ( $520 \text{ nm}$ )  $\mu\text{m}^{-2}$  in lower panel. Reprinted with permission from Nakatani K, Yau K-W. *J Physiol.* 1989;409:525–548.

ubiquitous in cells, regardless of whether the stimulus is light, odorants, hormones, neurotransmitters, etc. Even visual pigments bear a striking structural similarity to other G protein-coupled membrane receptors, all having a signature seven-transmembrane-helix topology. The knowledge of phototransduction is so advanced that it often serves as a model system to shed light on other signal transduction pathways.

As it is, the phototransduction process seems already well unfolded. However, like any complex cellu-

lar process, new components and side pathways will undoubtedly emerge with further probing. Even now, while most ingredients of phototransduction appear to have fallen into place, there are pieces of the puzzle that are either still at large (such as the  $\text{Ca}^{2+}$ -dependent regulator of the guanylate cyclase mentioned earlier) or in search of a well-characterized function (such as protein kinase C,<sup>162,163</sup> the protein phosphatase,<sup>188</sup> and so on). At the same time, while cones appear to have a similar phototransduction scheme as rods, the exact mechanisms underlying their low sensitivity and fast response kinetics still remain unclear. Other more subtle differences between rods and cones may also become evident upon closer examination. Most of the known proteins in the rod cascade are now cloned. This structural information, combined with the detailed biochemical and physiological knowledge, sets the stage for a molecular dissection of the transduction mechanism (see 32 for a recent review). One beneficiary of such advances will certainly be the understanding of genetic diseases affecting retinal photoreceptors. Already, the underlying causes of some of these diseases have come to light based on existing knowledge (see, for example, 189–191). Finally, one hopes eventually to be able to describe the entire phototransduction process in a quantitative way. This will not be an easy task because of the complexity of the process, but good progress is already being made in this direction.<sup>31,192–195</sup> The usefulness of a quantitative description is that it may often be the only practical way to evaluate the relative importance of different components or branches in a complex pathway.



**FIGURE 17.** A schematic representation of the phototransduction cascade (with the  $\text{Ca}^{2+}$  feedback not included) to highlight the importance of the decay time constants of the active intermediates ( $\tau_1$ ,  $\tau_2$ , and  $\tau_3$ ) and the basal rate of metabolic flux of cGMP in darkness ( $\tau_4$ ). G is transducin, and  $\text{PDE}_i$  is the inactive form of the phosphodiesterase. Asterisks indicate the active intermediates.  $\Delta\text{G} = 0$  represents the steady basal level of cGMP in darkness, and  $\Delta\text{G} < 0$  indicates a decline in cGMP level. Overall gain of the cascade (and hence light sensitivity) depends on the multiplicative product of the various  $\tau$  values.

### Acknowledgment

I thank Y. Koutalos for comments on the manuscript.

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